

Self-Modeling Machines, in Language

An LLM orchestrator that designs a real robot to grab, pull, and throw, then corrects its own model of it.

— PREMISE

Since Bongard, Zykov & Lipson (2006), a robot's *self-model*, its internal account of what its body is and can do, has been learned numerically from motor-babbling. It works, but a human cannot read it, and nobody can inspect it before the robot acts. We make the self-model linguistic. A **generator** LLM writes a structured, readable self-model of a VEX V5 manipulator from a finite, typed parts vocabulary: a morphology graph, a capability envelope (grip force, tractive load, release velocity), and a predicted score on a target task. A panel of **critic** LLMs attacks that self-model before a bolt is turned. Then comes the step that earns the word *self*. Once built, the robot reads its own body (VEX auto-detected devices, smart-motor encoders, the brain's IMU) and reconciles its **measured** self against the **imagined** one, revising the model in prose a human can audit. The body that models and the body modeled are the same object, exactly as in Lipson's work. What differs is that the representation is language, and it improves without retraining.

— THE LOOP

generator LLM → self-model {structure · capability · prediction} → **critic panel** attacks pre-build → build → **robot reads its own body** (devices · encoders · IMU · vision) → **reconcile** predicted vs measured → residual model corrects → repeat

— TWO LOOPS

LOOP 1 · CALIBRATION · THE CORE

One body, many cycles

Each task attempt yields proprioceptive and outcome data, and the self-model's predicted capabilities converge on the measured ones. It runs dozens of times a day without a rebuild, which is where the thesis is proven.

LOOP 2 · MORPHOLOGY · STRETCH, CUT FIRST

A few new shapes

We reconfigure the body two or three times, and each shape is re-authored and re-reconciled. This adds a cross-body story, though the result does not depend on it.

— GRAB, PULL, AND THROW

PRIMITIVE	SELF-MODEL PREDICTS	ROLE
Grab	max secure-hold (grip force)	prerequisite & safety net
Pull	max tractive work (load × distance)	calibration spine , the primitive that moves things
Throw	launch velocity → range	hero demo, sharpest physics

Pull is the spine because it carries low build-risk, stays continuously measurable, and does real mechanical work. Throw is the showpiece, and grab is the safety net. The three also compose one arc: grab the object, pull it into position, then throw it to a target, which gives three capability predictions on a single object.

— HARD PARTS AND RISKS

Signal quality. The LLM can only revise the self-model if VEX encoder and IMU telemetry separates model error from hardware variance. We log raw traces from the first build and verify SNR on pull force before scaling the loop.

Throw reliability. Precise release needs sub-10ms motor timing and a structure that survives repeated launches. If it slips, pull carries the calibration spine alone and throw becomes a recorded stretch goal.

Critic tuning. A panel that rejects everything stalls progress, and one that approves everything adds nothing, so calibrating critic strictness becomes its own tracked deliverable.

Throughput. Single-body trials run fast, but a plateau that only a rebuild can break costs hours, so one mid-project reconfiguration checkpoint tests for it early.

— WHY VEX V5

Reconciliation needs a body that can report itself, and VEX V5 can: 21 Smart Ports auto-detect every device, smart motors close the loop on encoders, the brain carries an IMU, and a native AI Vision Sensor reads AprilTags and objects. Its metal structure also survives the forces grab, pull, and throw demand. LEGO SPIKE's friction fits come apart under those loads, and RoFICoM only docks modules together, so it cannot manipulate a separate object. Among the surveyed platforms, VEX is the only one that is both a credible manipulator and a credible proprioceptor.

— THE ML AND LLM SPLIT

The **LLM** does the symbolic work: authoring the self-model, reconciling imagined against measured in language, and reasoning about why a prediction missed. A classical-ML **residual regressor** does the numeric work, learning from accumulating telemetry the systematic gap between predicted and actual as a function of commanded action, load, and battery voltage. The LLM cannot reliably know that the gripper loses force at low charge, but a regression over a few hundred trials learns exactly that. Because the two are separable, removing the regressor gives a clean ablation.

— WHY IT IS NOVEL

Numerical self-modeling Bongard 2006; Kwiatkowski & Lipson 2019	Builds self-models from proprioception, but they stay sub-symbolic, illegible, and never critiqued before the robot acts.
LLM morphology generation RoboMorph, 2024	An LLM works over a structured grammar, but it generates designs rather than self-models , runs only in simulation , and has no critic loop .
LLM self-improvement SAS-Prompt, ICRA 2025	Optimizes control parameters on real hardware, but it carries no morphology and no self-model .

No prior work reconciles an LLM-authored, language-grounded self-model against a robot's own proprioception, with adversarial critique before fabrication and numeric correction after it. **The contribution is that reconciliation step, where the imagined self is checked against the measured self in auditable language.**

— THE EXPERIMENT

Hypothesis. Self-model calibration error (predicted – measured capability) shrinks across Loop 1 cycles; the critics and the residual regressor each reduce it.

Conditions. (A) full loop · (B) no critics · (C) no residual regressor (LLM-only) · (D) human-designed control.

Metrics. calibration error per cycle (primary) · task success rate · failed predictions caught pre-build · cycles-to-target.

Primary result. the calibration curve, which is the robot's self-knowledge made measurable.

It satisfies two of the brief's directions at once: classical **ML combined with an LLM**, and **a different kind of agent** that models itself and explains its own failures in language.

— THE JUN 29 DEMO

A VEX V5 manipulator on a table, with a screen beside it showing the live self-model: predicted pull force, predicted throw range, grip confidence. Three beats.

Pull. The robot drags a known mass; predicted force and the encoder reading sit side by side; the LLM revises the model out loud as the gap closes.

Throw. Predicted range shows before release; the object lands; the model updates with the measured distance, and the critic rejection log explains any catch.

The curve. Predicted-versus-actual error across every Loop 1 trial, trending down. That is the result: the robot's self-knowledge, measured.

References. [1] Bongard, Zykov & Lipson. Resilient Machines Through Continuous Self-Modeling. *Science* 314(5802):1118–1121 (2006). DOI 10.1126/science.1133687. [2] Kwiatkowski & Lipson. Task-Agnostic Self-Modeling Machines. *Science Robotics* 4(26):eaau9354 (2019). DOI 10.1126/scirobotics.aau9354. [3] Lipson & Pollack. Automatic Design and Manufacture of Robotic Lifeforms. *Nature* 406:974–978 (2000). [4] Qiu et al. RoboMorph: Evolving Robot Morphology using Large Language Models. arXiv:2407.08626 (2024). [5] Ben Amor et al. SAS-Prompt: LLMs as Numerical Optimizers for Robot Self-Improvement. ICRA 2025. arXiv:2504.20459. [6] Eye-in-Finger: Smart Fingers for Delicate Assembly and Disassembly of LEGO. arXiv:2503.06848 (2025).